

Dr. Babasaheb Ambedkar Technological University Lonere - Raigad

Semester Examination: ~~December 2014~~ May 2015

Class: B. Tech. Computer Engineering / Information Technology

Semester: V

Subject: Database Management System

Time: 3 hours

Date: ~~25/05/15~~ 22/05/15

Maximum Marks: 70

- Instructions: 1. Question 1 is compulsory.
2. Attempt Any one question from Group I, II, III, IV.
3. Attempt one question from remaining.

- Q.1. Solve the following (2 Mark Each) : (10)
- A. In an E-R diagram double lines indicate
(A) Total participation. (B) Multiple participation.
(C) Cardinality N. (D) None of the above.
- B. The operation which is not considered a basic operation of relational algebra is
(A) Join. (B) Selection. (C) Union. (D) Cross product.
- C. When an E-R diagram is mapped to tables, the representation is redundant for
(A) weak entity sets (B) weak relationship sets
(C) strong entity sets (D) strong relationship sets
- D. If the closure of an attribute set is the entire relation then the attribute set is a
(A) superkey (B) candidate key (C) primary key (D) not a key
- E. If two relations R and S are joined, then the non matching tuples of both R and S are ignored in
(A) left outer join (B) right outer join
(C) full outer join (D) inner join

Group - I

- Q.2. A. Write an java function using JDBC metadata features that takes ResultSet as an input parameter, and prints out the result in tabular form, with appropriate names as column headings. (06)
- B. Consider the relational database above, where the primary keys are underlined. Give an expression in the tuple relational calculus for each of the following queries: (06)
- a. Find all employees who work directly for "Jones".
b. Find all cities of residence of all employees who work directly for "Jones".
c. Find the name of the manager of the manager of "Jones".
- Q.3. A. Give the syntax, purpose and example of the following functions : (06)
1. COUNT(*) 2. ROUND 3. INITCAP
4. MIN 5. LENGTH 6. AVG

- B. Write the following queries in relational algebra, using the university schema. (06)
- Find the titles of courses in the Comp.Sci department that have 3 credits.
 - Find the highest salary of any instructor.
 - Find the IDs of all students who were taught by an instructor named Einstein; make sure there are no duplicates in the result.

Group - II

- Q.4. A. Which key is used for a dependent entity, a Non-composite or a Composite key? (06)
How is a dependent entity represented in an E-R diagram? If *Teacher* entity depends on *Department* entity how will you represent it in an E-R diagram?

- B. Suppose there is table of the following form: (06)

A#	Aname	BooksWritten
100	Mehta	MS SQL Server
100	Mehta	Sybase
110	Nanda	Informix
110	Nanda	Oracle

What kind of relationship is represented – One-to-one, one-to-many, many-to-one or many-to-many? How will you break this into multiple tables? Explain

- Q.5. A. By using Aggregation or Having clause, write the query for the following (06)
- Find the average salary in each department.
 - Find the average salary of instructors in those departments where the salary is more than \$42,000.

- B. Write the SQL queries for the following: (06)
- Create a view that lists all course sections offered by the Physics department in the Fall 2009 semester with the building and room number of each section.
 - Show how to define a view *tot_credits (year, num_credits)*, giving the total number of credits taken by students in each year.

Group - III

- Q.6. A. Which factors should to be taken into consideration while choosing a RAID level? (06)

- B. Compare BCNF and 3NF with suitable example. (06)

- Q.7. A. Explain with a diagram (06)
- Shared Nothing Parallel database architecture
 - Hierarchical Parallel database architecture

- B. Consider the following relations with primary key underlined (06)
- Customer (C#, Cname, Address)
 Item (I#, Iname, Price, Weight)
 Order (O#, C#, I#, Quantity)
 Write SQL queries for the following:

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- a. List the names of customers who have ordered items weighing more than 1000 and only those.
- b. List the names of customers who have ordered atleast one item priced over Rs.500.
- c. Create a view called "orders" that has the total cost of every order.

Group - IV

Q.8. A. Define the following related to indices: (06)

- a. Access types b. Access time c. Insertion time
- d. Deletion time e. Space overhead f. Search key

B. Explain basic steps involved in Query processing with the help of block diagram. (06)

Q.9. A. What is mean by Ordered Indices? Explain the structure of Dense and Sparse Index. (06)

B. Explain the Server System Architecture of a database system. (06)

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